AWS VPC: Quick Revision Notes

Short notes: definitions, parts, CIDR, public vs private, IGW, NAT, peering, security and the key steps

1. VPC in One Minute

- **VPC** = Virtual Private Cloud: your own **logically isolated network** in AWS. You choose the IP range, subnets, route tables and gateways.
- A VPC lives in **one region** and spans **all AZs** of that region. A **subnet** lives in **one AZ**.
- **Default VPC:** made by AWS in every region: 172.31.0.0/16, one public /20 subnet per AZ, IGW attached. **Custom VPC:** you build it (for example 10.0.0.0/16).

Picture: VPC = your gated compound in the AWS city. Subnets = blocks inside it. Route tables = roads. Gateways = gates.

2. Parts of a VPC

Part	What it does
Subnet	Piece of the VPC range, in one AZ. Public or private.
Route table	Says where traffic goes. Each subnet is associated with one.
Internet Gateway (IGW)	Door between the VPC and the Internet. Free.
NAT Gateway	Private subnet can go **out** to the Internet; nobody can come **in**. In a public subnet, needs an Elastic IP, costs money.
Security group	Instance-level firewall. Stateful. Allow rules only.
Network ACL	Subnet-level firewall. Stateless. Allow and deny rules.
Elastic IP	Fixed public IPv4 address.
VPC peering	Private link between two VPCs (non-overlapping CIDRs, not transitive).
VPN / Direct Connect	Private link to your office or data center.

3. CIDR and Reserved IPs

Item	Remember
Total addresses	$2^{(32 - prefix)}$. /16 = 65,536, /24 = 256, /28 = 16
Usable in AWS	Total **- 5**. A /24 subnet has **251** (not 254)
VPC / subnet size	Between **/16** and **/28**
Reserved in every subnet	**.0** network, **.1** VPC router, **.2** Amazon DNS, **.3** future use, **last** broadcast
Split a block	Add n bits so that 2^n is at least the number of subnets. 10.0.0.0/16 into 4 subnets = **/18**: 10.0.0.0, 10.0.64.0, 10.0.128.0, 10.0.192.0
Private ranges	10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16

```
Subnet 10.0.0.0/24
10.0.0.0 network | 10.0.0.1 router | 10.0.0.2 DNS | 10.0.0.3 future
10.0.0.4 to 10.0.0.254 usable (251) | 10.0.0.255 broadcast
```

4. Public vs Private Subnet

	Public subnet	Private subnet
Route table	0.0.0.0/0 to the **Internet Gateway**	No IGW route (0.0.0.0/0 to a **NAT Gateway** for outbound only)
Instance IP	Public IP (auto-assign or Elastic IP)	Private IP only
From the Internet	Reachable (if the security group allows)	Not reachable
Typical use	Web servers, load balancers, bastion	Databases, app servers

Remember: A subnet is **public** only because its route table points to an IGW. The name does not matter.

5. Key Steps at a Glance

Make a public subnet

Step	Action
1	Create VPC: name MyVPC, CIDR 10.0.0.0/16, enable DNS hostnames.
2	Create subnet: PublicSubnet, one AZ, 10.0.0.0/24.
3	Create Internet Gateway MyIGW and **attach** it to MyVPC.
4	Create route table PublicRT: add **0.0.0.0/0 to MyIGW**.
5	**Associate** PublicSubnet with PublicRT.
6	Enable **auto-assign public IPv4** on the subnet.
7	Launch an instance in PublicSubnet, security group SSH from My IP.

Give a private subnet outbound Internet (NAT)

Step	Action
1	Create PrivateSubnet (10.0.1.0/24) and PrivateRT associated with it. No IGW route.
2	Create NAT Gateway in the **public** subnet, allocate an Elastic IP. Wait for Available.
3	PrivateRT: add **0.0.0.0/0 to the NAT Gateway**.
4	Test from the private instance: `curl -s ifconfig.me` shows the NAT's Elastic IP.

Connect two VPCs (peering)

Step	Action
1	Make sure the CIDRs do **not overlap** (10.0.0.0/16 and 10.1.0.0/16).
2	Peering connections, Create: requester VPC and accepter VPC.
3	Accepter: Actions, **Accept request** (status Active).
4	Add routes in **both** VPCs: destination = the other VPC's CIDR, target = the peering connection.
5	Allow the traffic in the **security groups**.

6. Security Group vs Network ACL

	Security group	Network ACL
Level	Instance	Subnet
Rules	Allow only	Allow and deny, in number order
Stateful?	Yes (replies allowed)	No (allow replies too)
Default	Inbound blocked, outbound allowed	Default NACL allows all

7. Commands to Remember

Command (on an instance)	Shows
`ip -4 addr show`	Private IP inside the subnet (starts at .4 or higher)
`ip route`	default via .1 = VPC router
`curl -s ifconfig.me`	Public IP (the NAT's Elastic IP from a private instance)
`ping -c 2 8.8.8.8`	Internet reachable?
`ssh -i key.pem -J ubuntu@PUBLIC_IP ubuntu@PRIVATE_IP`	Reach a private instance through the public one

8. Quick Questions

Question	Answer
VPC is region or AZ based? Subnet?	VPC: region. Subnet: one AZ.
How many IPs does AWS reserve per subnet?	5. A /24 has 251 usable.
What makes a subnet public?	A route 0.0.0.0/0 to an Internet Gateway.
What does a NAT Gateway do?	Lets private instances go out to the Internet, not in.
Where is the NAT Gateway placed?	In a public subnet, with an Elastic IP.
Is VPC peering transitive?	No.
Peering needs what?	Non-overlapping CIDRs, routes in both VPCs, security group rules.
Security group vs NACL?	Instance, stateful, allow only / subnet, stateless, allow and deny.